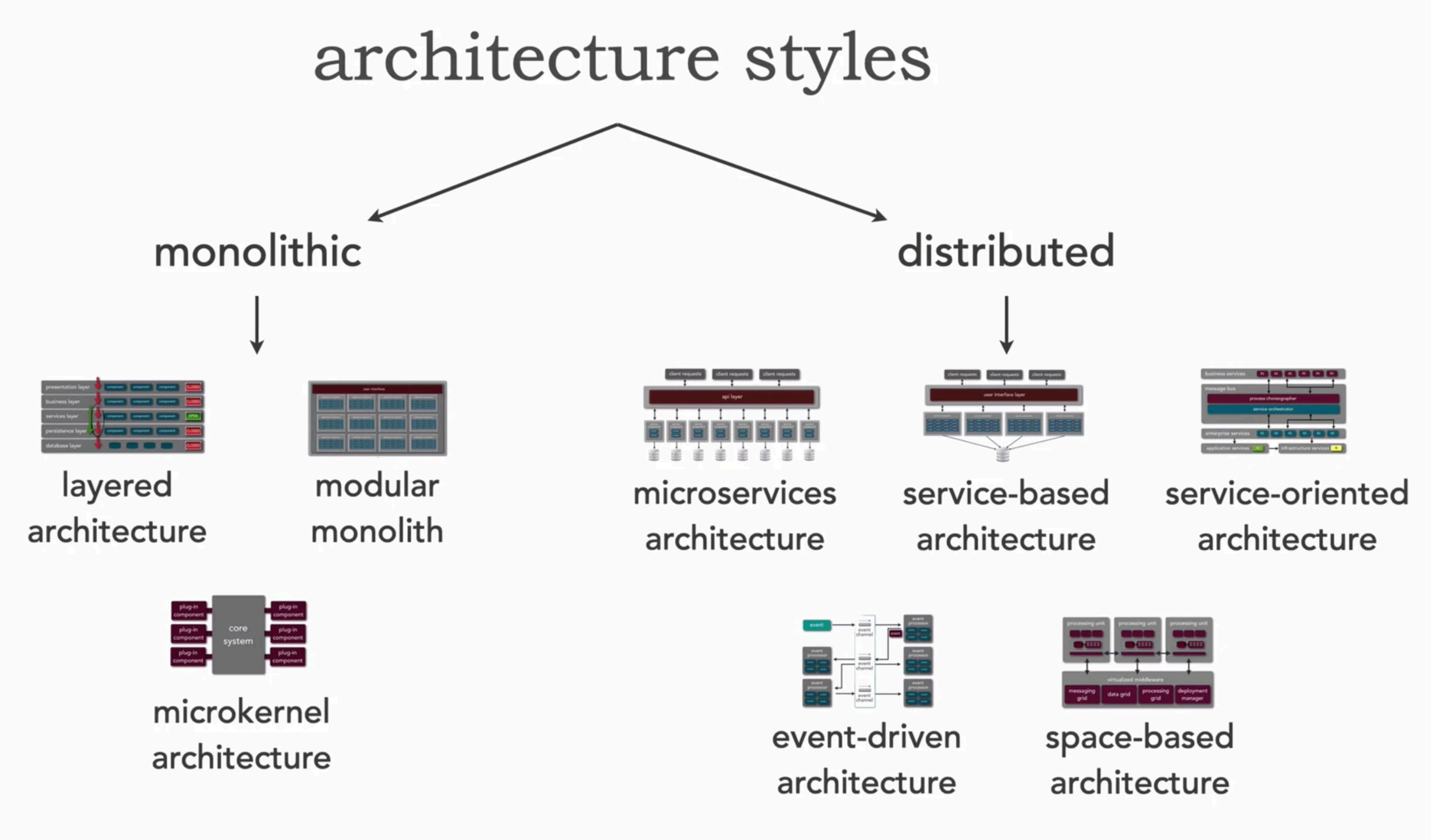
Software Architectures

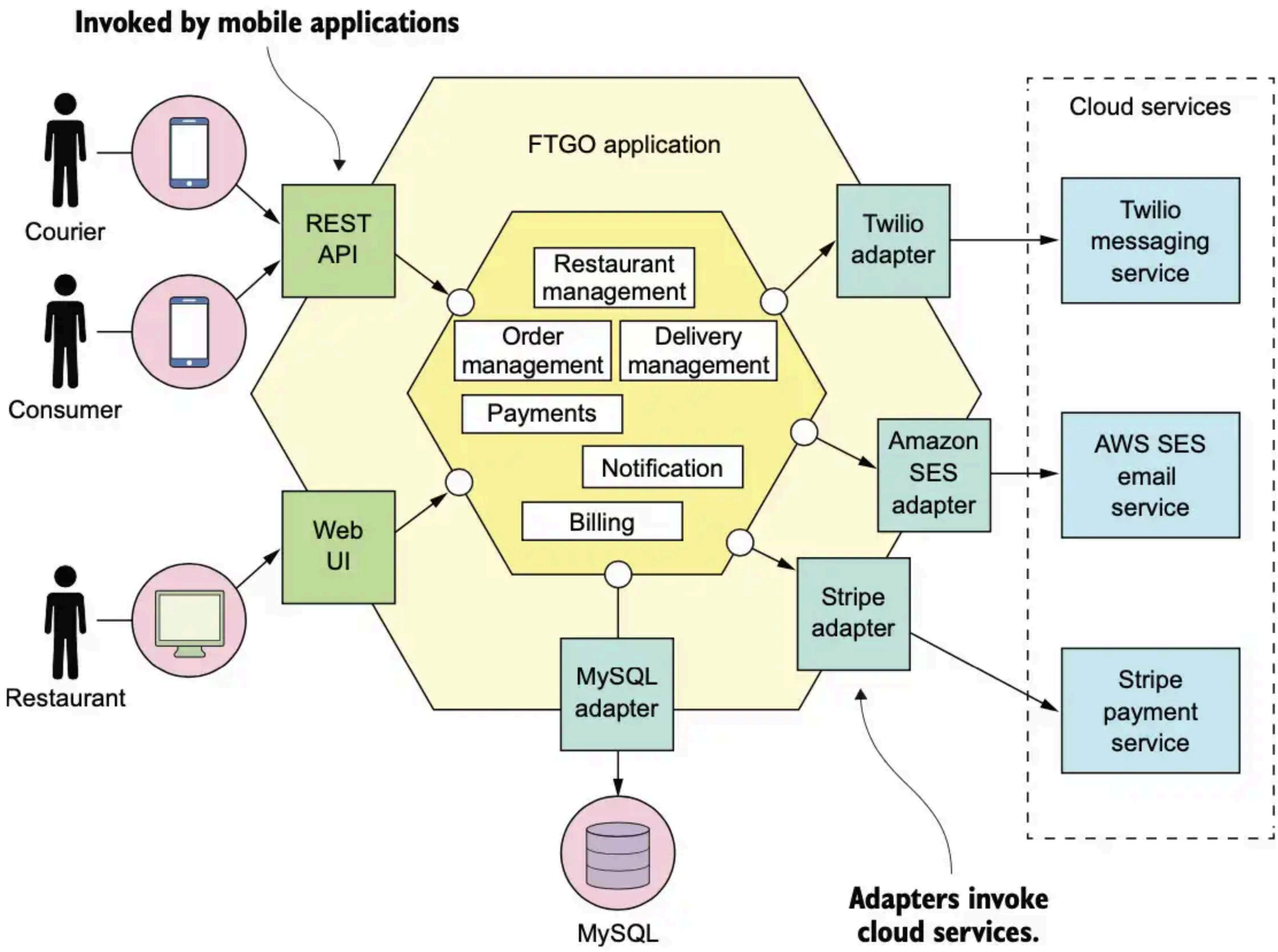
What is Software Architecture?

Architecture defines **components, connectors, and constraints** that shape how systems are organized. Different styles change **local complexity** (inside modules) and **global complexity** (system-wide).



Monolithic Architectures

A **single, unified application** where all parts run in one process.

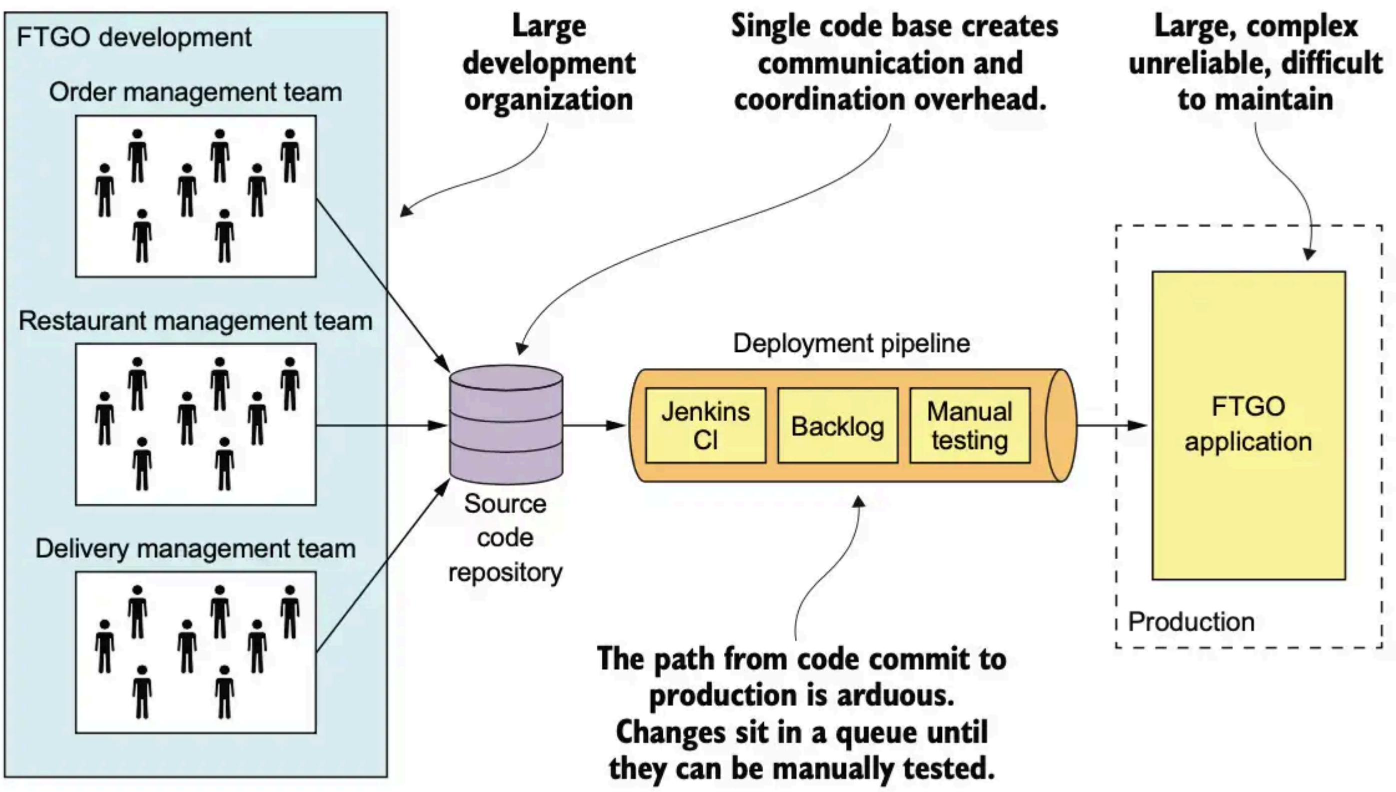


Pros

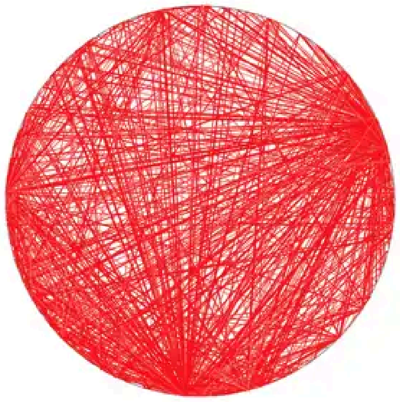
- Simple development, testing, deployment
- Fast in-process communication

Cons (when project grows)

- Tends toward a *big ball of mud*
- Slow development loops (startup, testing, deployment)
- Hard to refactor
- Tech lock-in
- Can't scale or deploy individual pieces

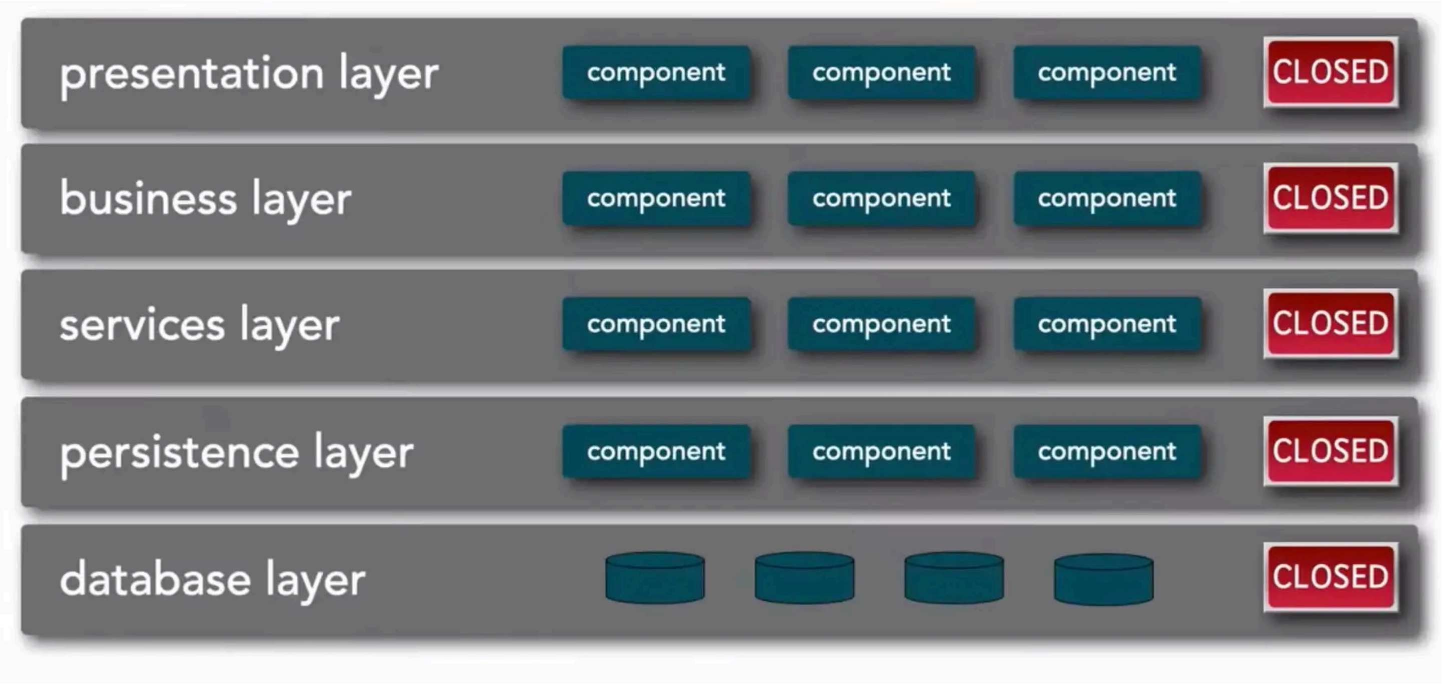


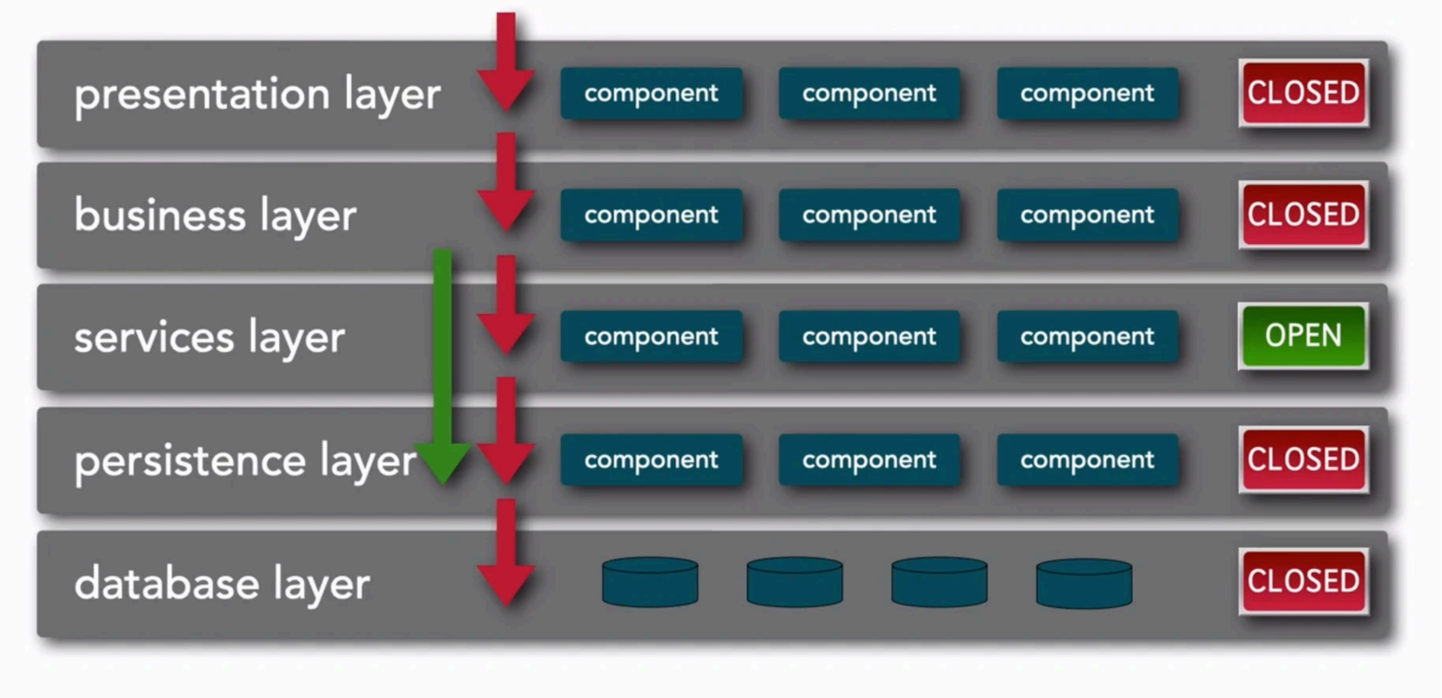
Big Ball of Mud



Multi-Layered (N-tier) Architecture

Application split into technical layers (UI → business → data).





Pros

- Clear separation of concerns
- Easy for teams to work on different layers
- Familiar and well-supported

Cons

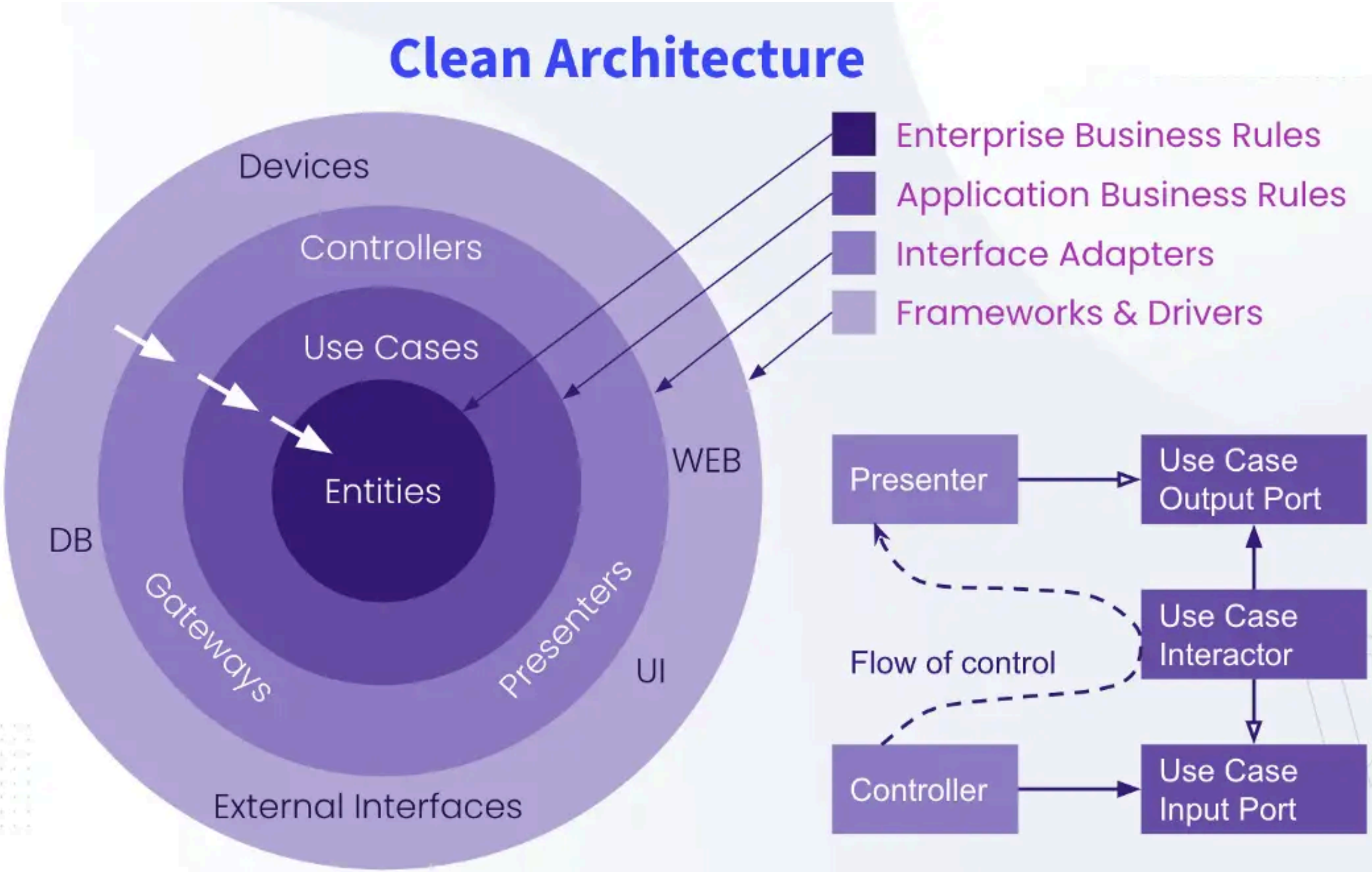
- Vertical features require changes across layers
- Inefficient message flow through layers
- Harder to refactor around domain boundaries

Use

- Simple applications
- Technically structured teams

Clean Architecture

Domain and use-cases at the center; frameworks and infrastructure at the edges.



Pros

- Strong decoupling
- Easy to test business logic
- Swap DB/UI/frameworks without touching core logic

Cons

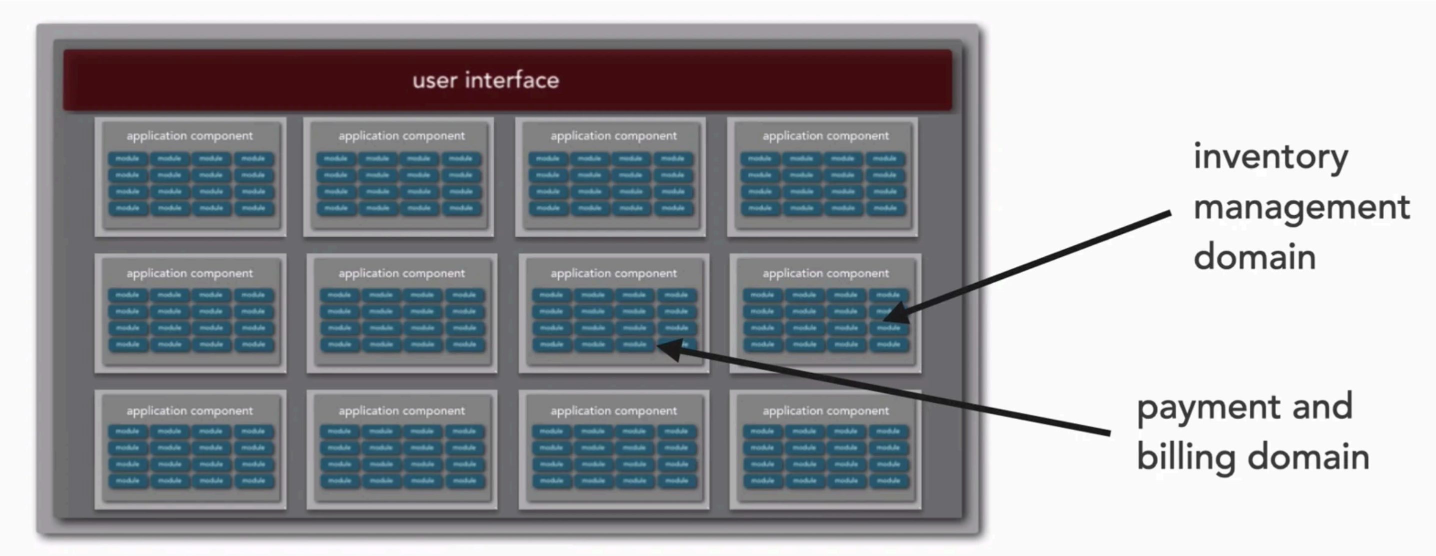
- More abstraction and indirection
- Requires architectural discipline

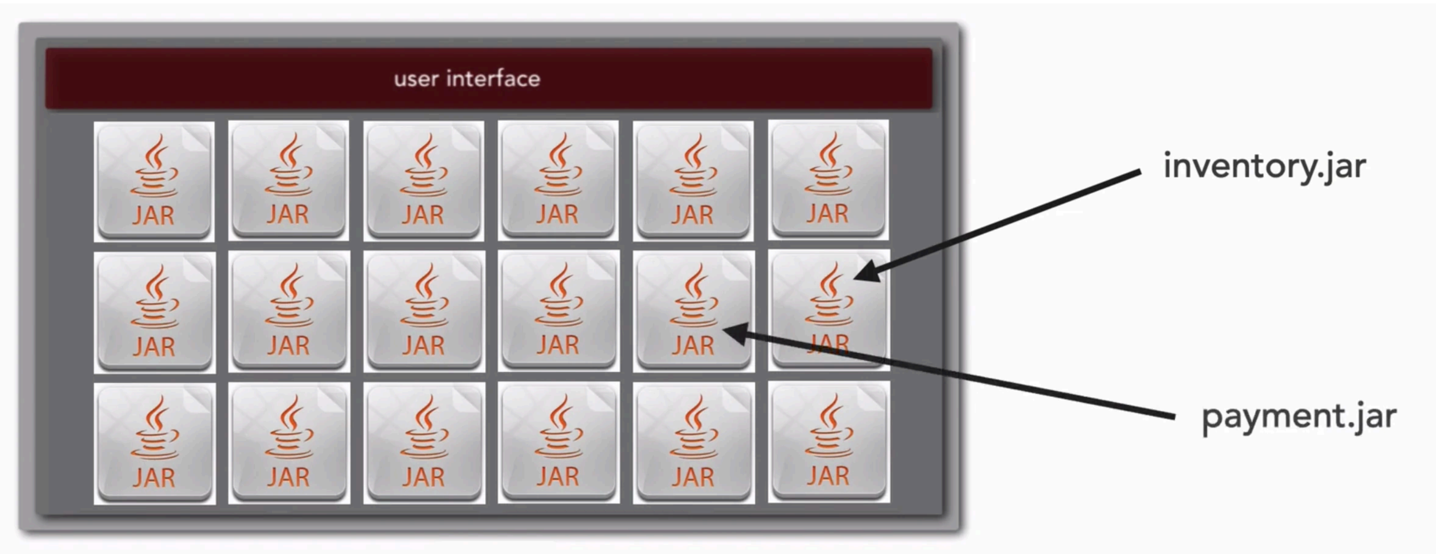
Use

- Systems expected to evolve for years
- Applications with complex business rules

Modular Monolithic Architecture

Organize code by **business modules**, not layers.





Pros

- Strong domain encapsulation
- Good consistency (single database)
- Natural path toward microservices

Cons

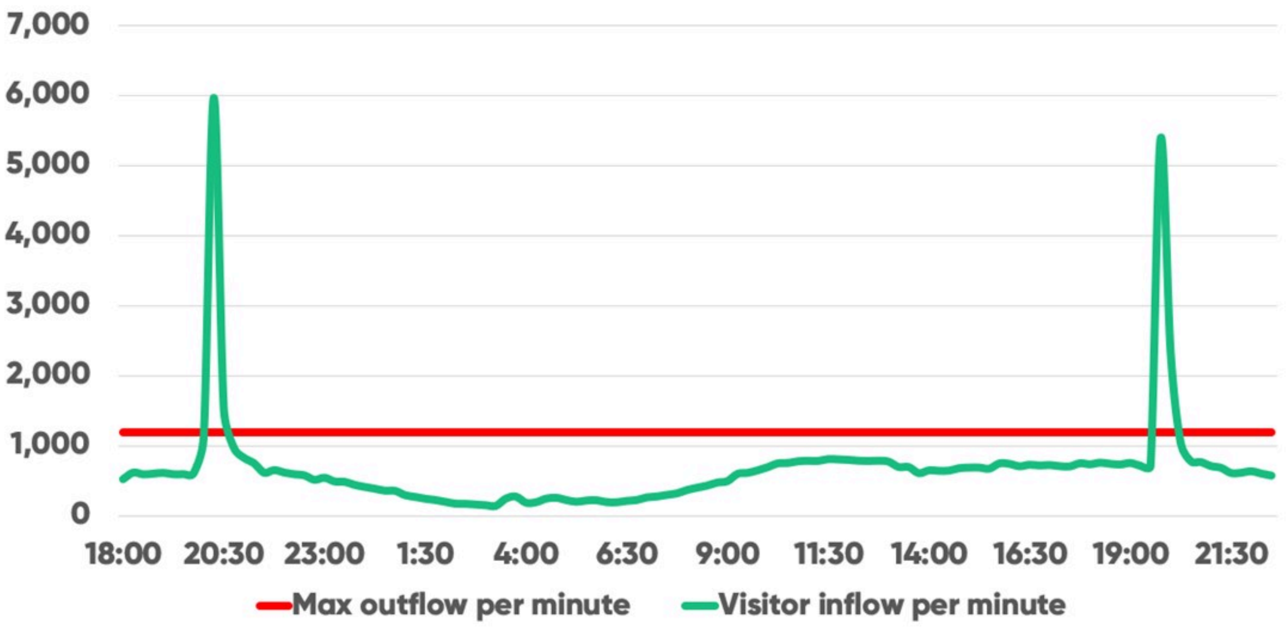
- No independent deployment
- No technology polyglot
- Scaling still applies to entire app

Use

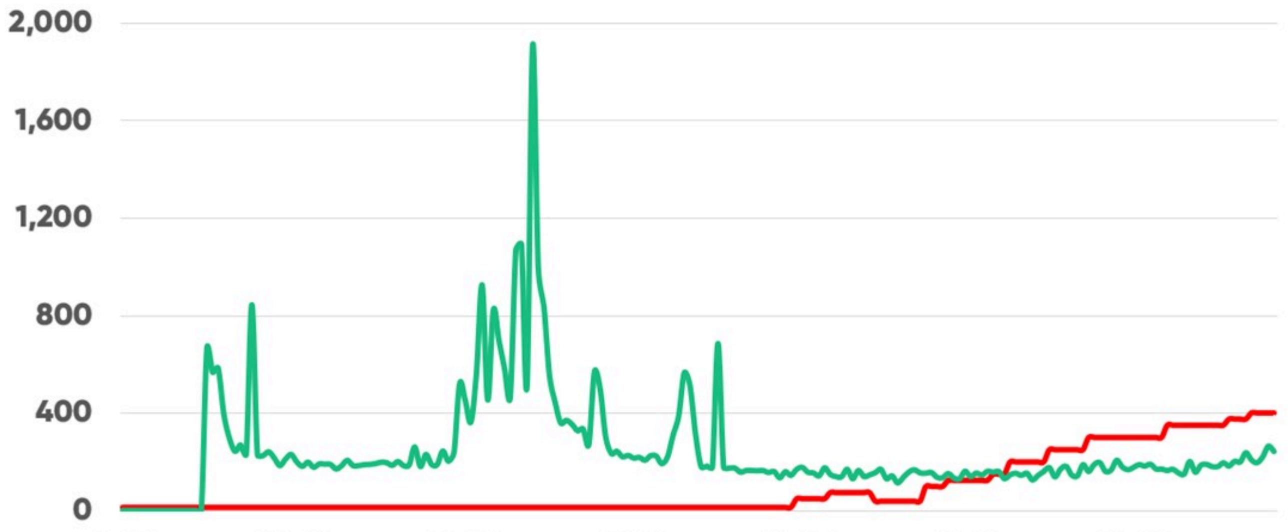
- Domain-oriented teams
- Large monoliths undergoing modernization
- Workloads requiring strong consistency

Monoliths and the Challenge of Traffic Spikes

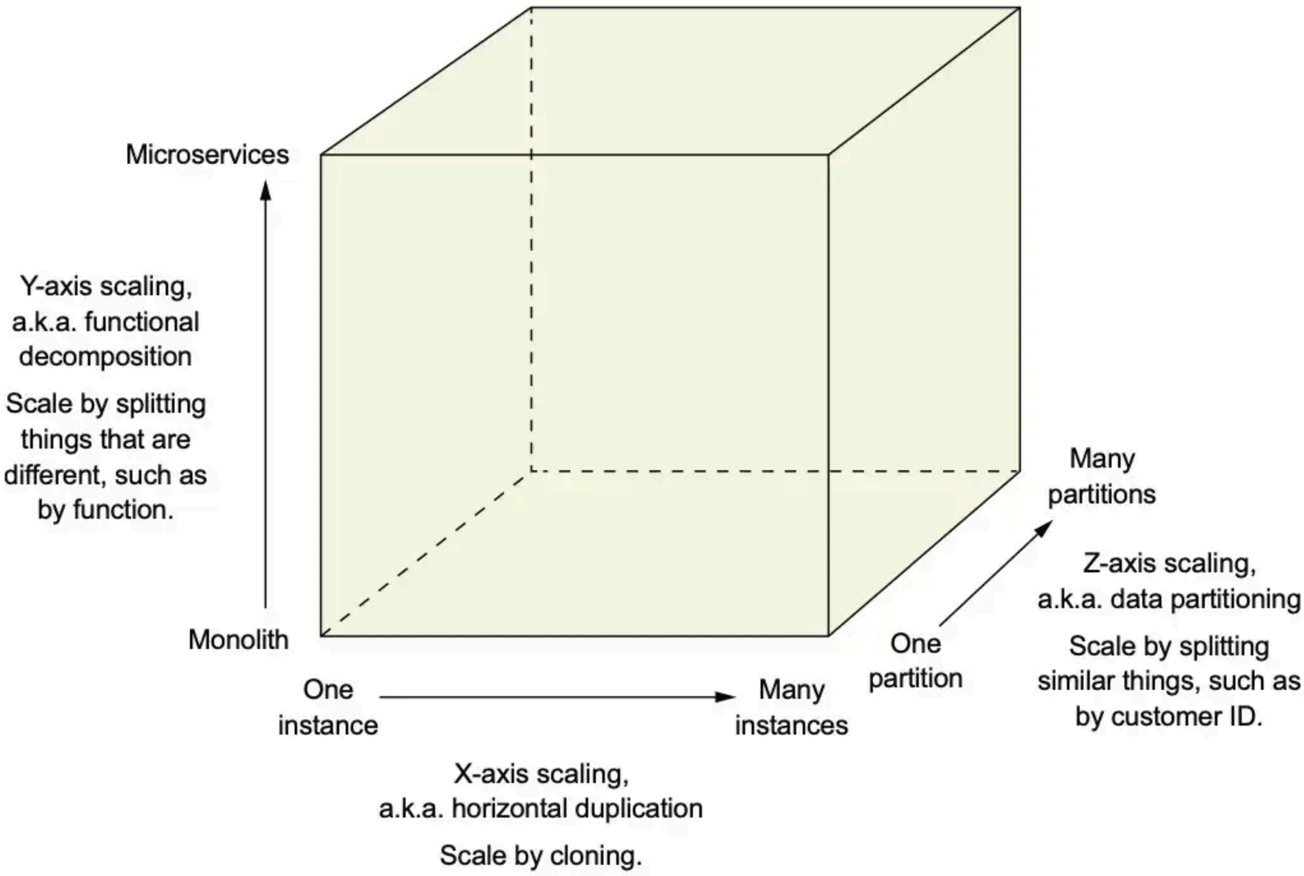
PR appearance traffic spike: Rakuten France



Exclusive product drop traffic spike: Rapha

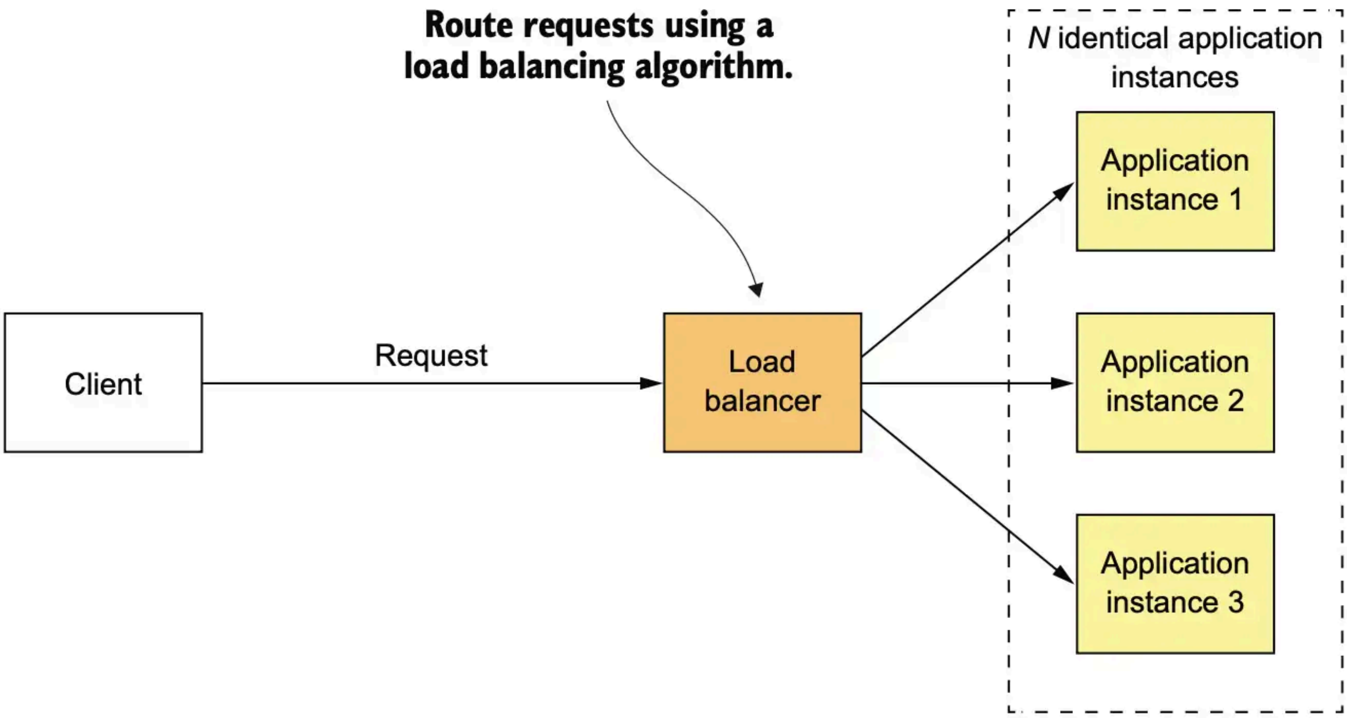


Scaling Monoliths — The Scale Cube



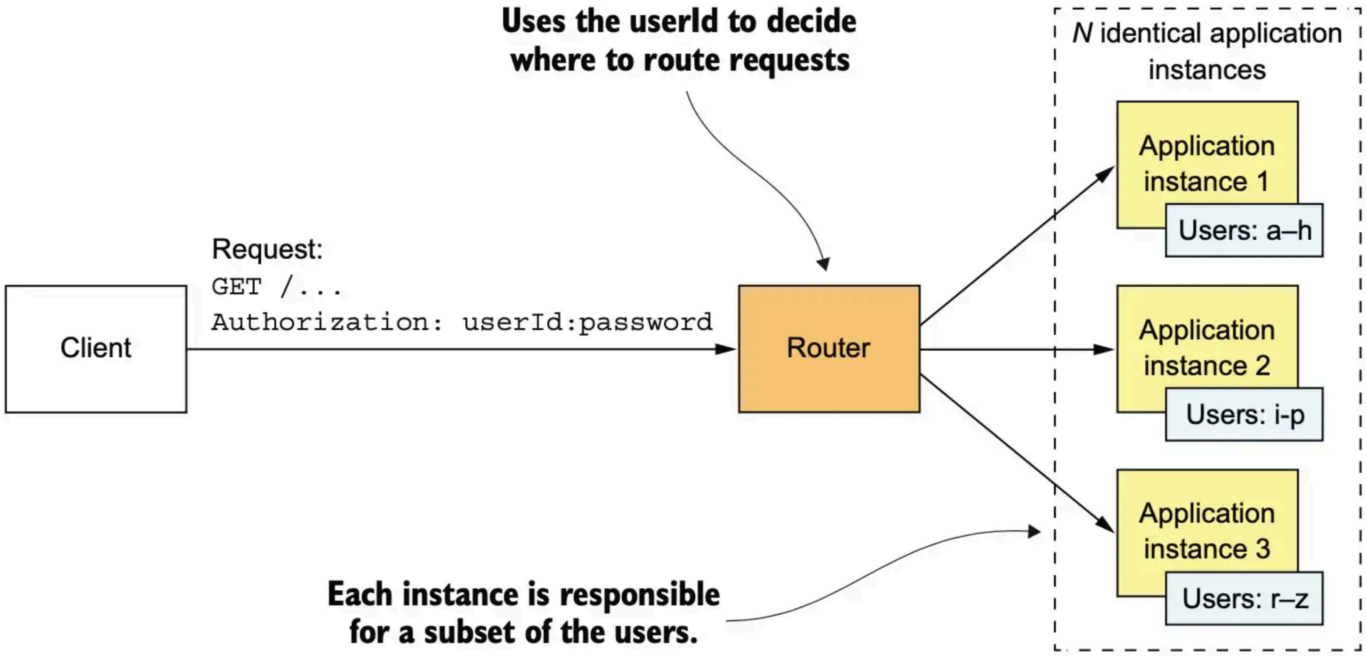
X-axis Scaling

Clone the entire application behind a load balancer.



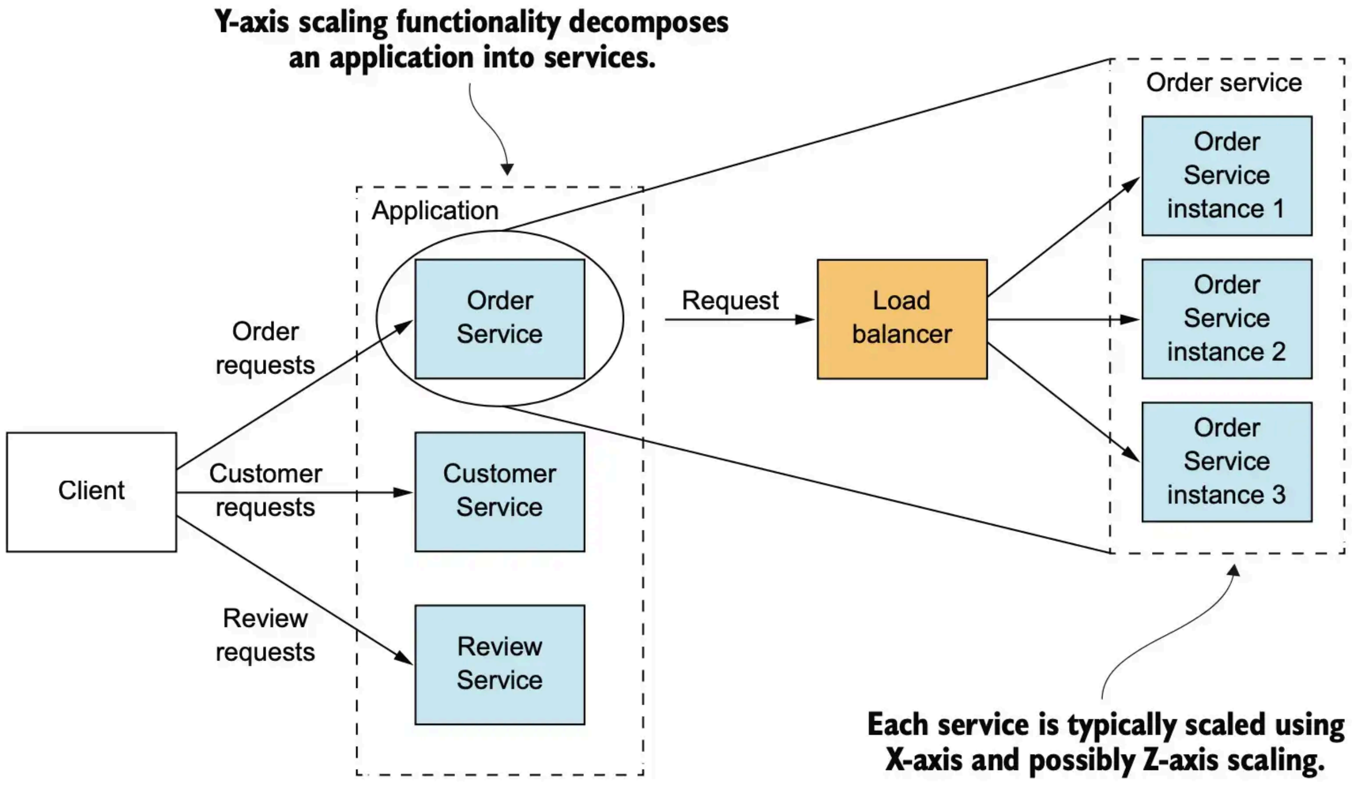
Z-axis Scaling

Shard by data (e.g., userId).



Y-axis Scaling

Split the application by function.



Distributed Architectures

Applications composed of multiple components running on different machines, communicating over RPC.

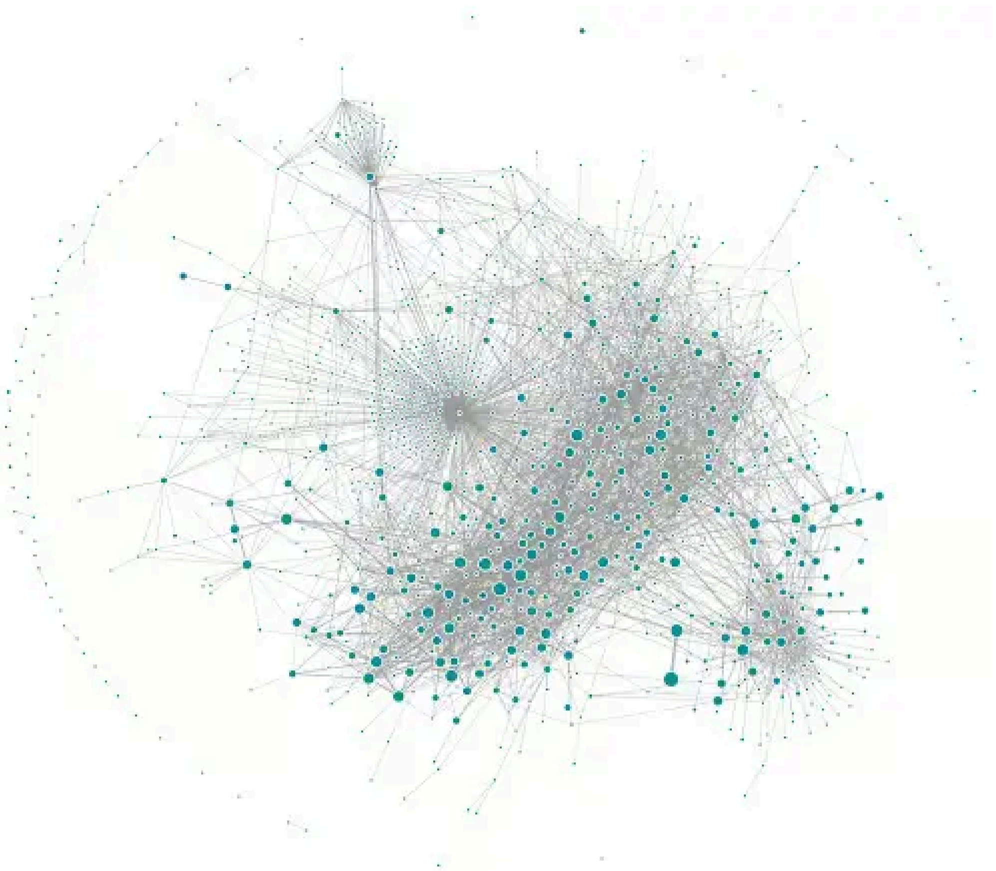
Pros

- Independent deployment

- Independent scaling
- Smaller units → lower local complexity

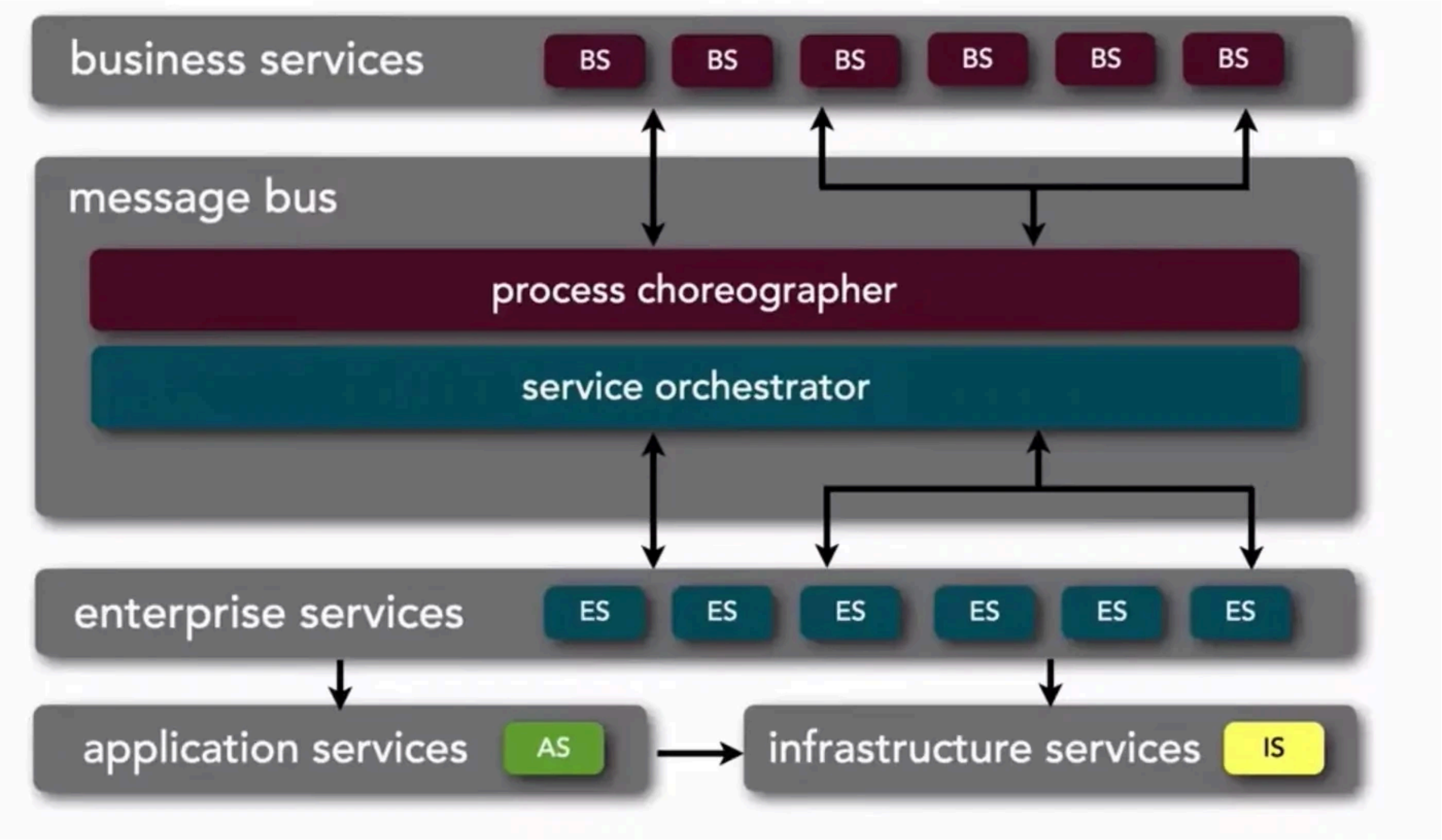
Cons

- High global complexity (APIs, retries, consistency)
- Requires service discovery, tracing, logging
- Risk of a *distributed monolith*



Service-Oriented Architecture (SOA)

Dumb services, smart pipes. Often built around shared enterprise services and protocols like SOAP.



Pros

- Good for integrating heterogeneous systems
- Reusable, discoverable services

Cons

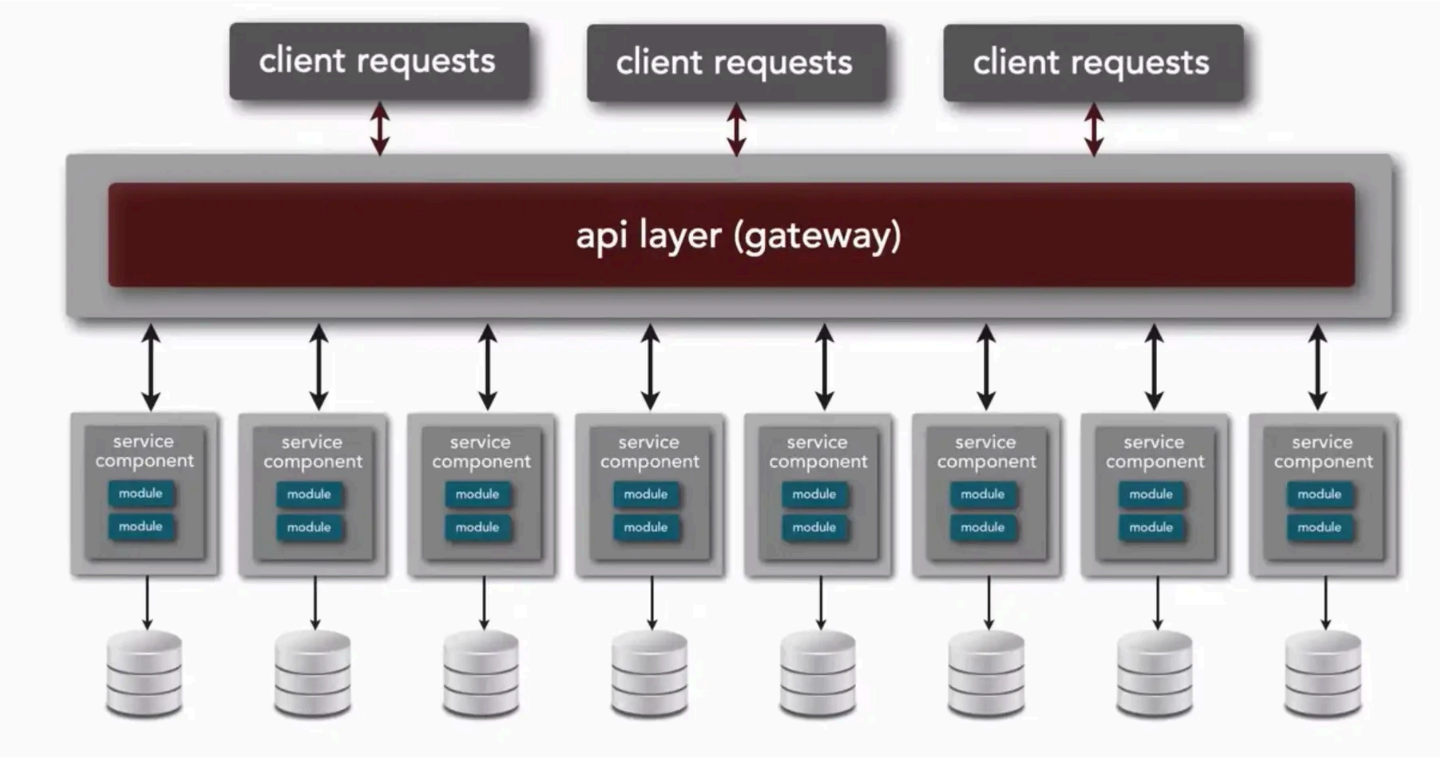
- Heavy protocols
- Shared services become bottlenecks
- Slow adaptation to change

Use

- Large enterprises with legacy systems
- Cross-system workflows

Microservices Architecture

Smart services, dumb pipes. Each service owns its data and is deployed independently.



Pros

- High evolvability
- Independent deploy and scale
- Technology freedom
- Team autonomy

Cons

- Operational complexity
- Harder debugging and monitoring
- Many small services/databases to manage

Use

- Complex systems with fast evolution
- Teams experienced in distributed systems

Resources

- *Fundamentals of Software Architecture* — Richards & Ford
- Mark Richards – YouTube channel